

Data Flow Diagram

A Comprehensive Measure of Well-Being | HDI Predictor

Level 0 - Context Diagram

User inputs development indicators → HDI Predictor System → Outputs HDI Score + Tier

Level 1 - Main Processes

- 1.0 Accept User Input (Country, Life Exp, MYS, EYS, GNI)
- 2.0 Validate Input Ranges
- 3.0 Compute Health Index = $(\text{life} - 20) / (85 - 20)$
- 4.0 Compute Education Index = $(\text{MYS}/15 + \text{EYS}/18) / 2$
- 5.0 Compute Income Index = $\log(\text{GNI})$ normalized
- 6.0 Compute HDI = geometric mean of 3 indices
- 7.0 Classify Tier and Display Result

Data Stores

- Country Presets (UNDP HDR 2023/24 - 20 countries)
- ML Model (model.pkl)
- Label Encoder (encoder.pkl)